

Lecture 11: Moment Generating Functions

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1 Moment generating functions::: {.callout-note icon=false}## Definition: MGF

$$M_X(t) = E[e^{tX}].$$

Differentiating and setting $t = 0$ recovers moments: $M'_X(0) = E[X]$, $M''_X(0) = E[X^2]$, etc....
. ::: {.callout-warning icon=false}## FactIf X, Y independent, $M_{X+Y}(t) = M_X(t)M_Y(t)$.
Two r.v.'s with the same MGF near 0 have the same distribution.:::—## MGF of a Normal, empirically 1

—title: “Lecture 11: Moment Generating Functions”author: “Jamie Haddock”format: revealjs: output-file: Lecture11_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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```
mu, sigma = 0, 1X = rng.normal(mu, sigma, size=200000)ts = np.linspace(-1, 1, 50)empirical_mgf = [np.me
```

—# Case Study: A Sub-Gaussian Bound for Rademacher Variables::: {.callout-warning icon=false}## FactA Rademacher $z \in \{-1, +1\}$ has $M_z(t) = \cosh(t) \leq e^{t^2/2}$ for all t – exactly the algebraic fact that proves concentration for the trace estimator in three lectures:::

```
ts = np.linspace(-3, 3, 200)plt.plot(ts, np.cosh(ts), label=r"$M_z(t)=\cosh(t)$")plt.plot(ts, np.exp(ts
```

Midterm 1 is Thursday, covering everything through today.## Reading & HomeworkReading: §4.5 (Bertsekas & Tsitsiklis, *Introduction to Probability*).Due today: Homework 5 (Gradescope).